StockSense — Component Shortlist & Technical Justifications

1. Microcontroller (MCU) Core

Component	Selected Option	Secondary / Candidate Option
Model	**ESP32-WROOM-32U** (External Antenna)	STM32F401 / STM32L4 (Future Low-Power Variant)
Architecture	32-bit Dual-Core Xtensa LX6 @ 240 MHz	32-bit ARM Cortex-M4 @ 84 MHz
Memory	520 KB SRAM, 4 MB / 8 MB SPI Flash	96 KB SRAM, 512 KB Flash
Wireless	Integrated Wi-Fi (802.11 b/g/n) & Bluetooth v4.2 BR/EDR + BLE	External Wi-Fi (ESP8266 / AT Commands via UART)

Technical Justification

- **Native Connectivity & TLS Support:** StockSense relies on MQTT over TLS (port 8883) for secure node-to-cloud communication. The ESP32 hardware cryptographic accelerators (AES, SHA-2, RSA) execute TLS handshakes and encryption without stalling real-time sensor tasks.
- **Dual-Core Architecture:** Core 0 handles low-level wireless stacks (Wi-Fi, TCP/IP, MQTT TLS), while Core 1 runs the FreeRTOS scheduler, digital signal filtering (Kalman filter for strain gauges), and RFID interrupt handlers.
- **Offline Storage Capabilities:** Integrated SPI Flash allows partition mapping for SPIFFS/LittleFS, enabling local Ring Buffer queuing during connectivity dropouts.
- **External Antenna (WROOM-32U):** Storage racks, metal shelf structures, and commercial storage rooms attenuate RF signals; an external U.FL rubber duck antenna ensures stable Wi-Fi performance compared to PCB trace antennas.

2. Analog Front-End (AFE) & Load Cell Sensor

Component	Selected Option	Secondary / Candidate Option
ADC IC	**HX711** (24-bit Sigma-Delta ADC)	ADS1232 (24-bit Low-Noise ADC)
Gain / Rates	On-chip PGA (Selectable Gain 64 / 128), 10 Hz / 80 Hz SPS	Programmable Gain (1, 2, 64, 128), 10 Hz / 80 Hz SPS
Load Cell Type	**Single-Point Parallel Beam Strain Gauge** (5 kg / 20 kg / 50 kg)	S-Type / Button Load Cells

Technical Justification

- **High Resolution for Weight-Based Counting:** StockSense targets a quantity accuracy of ±1%. The HX711 offers 24-bit resolution (over 16 million noise-free steps), rendering minute weight changes (e.g., removing a single 10 g item from a 20 kg load bin) easily detectable after digital filtering.
- **Differential Input Channels:** Channel A provides selectable gain (128 or 64) directly matched to Wheatstone bridge voltage outputs (2 mV/V nominal sensitivity). Channel B can be utilized for auxiliary system diagnostic monitoring.
- **Cost-to-Performance Ratio:** HX711 is widely accessible, low-cost, and easily integrated into custom PCB designs with minimal passive support circuitry.

3. Item Identification & Environmental Sensors

3.1 RFID Scanner Module

Parameter	Selected: MFRC522	Alternative: PN532
Frequency	13.56 MHz (HF)	13.56 MHz (HF)
Supported Protocols	ISO/IEC 14443 A / MIFARE	ISO/IEC 14443 A/B, FeliCa, NFC
Bus Interface	SPI (Up to 10 Mbit/s)	SPI / I^2C / High-Speed UART

Justification: The MFRC522 provides rapid tag scan times (<50 ms) over an SPI interface, consuming low standby power. It identifies individual bins, tagged items, or user ID authorization cards during stock additions or removals.

3.2 Climate Monitoring Sensor

Parameter	Selected: SHT30 / SHT31	Alternative: DHT22 / AM2302
Interface	I^2C (Selectable Addresses 0x44 / 0x45)	Single-Bus Custom Serial
Accuracy	±2% RH, ±0.2°C	±2–5% RH, ±0.5°C
Operating Range	0 to 100% RH, –40°C to +125°C	0 to 100% RH, –40°C to +80°C

Justification: SHT30 communicates over standard I^2C (shared with status displays), enabling clean bus topology on the custom PCB. Accurate humidity and temperature logging are vital for moisture-sensitive stock in pharmacy, clinic, or restaurant storage environments.

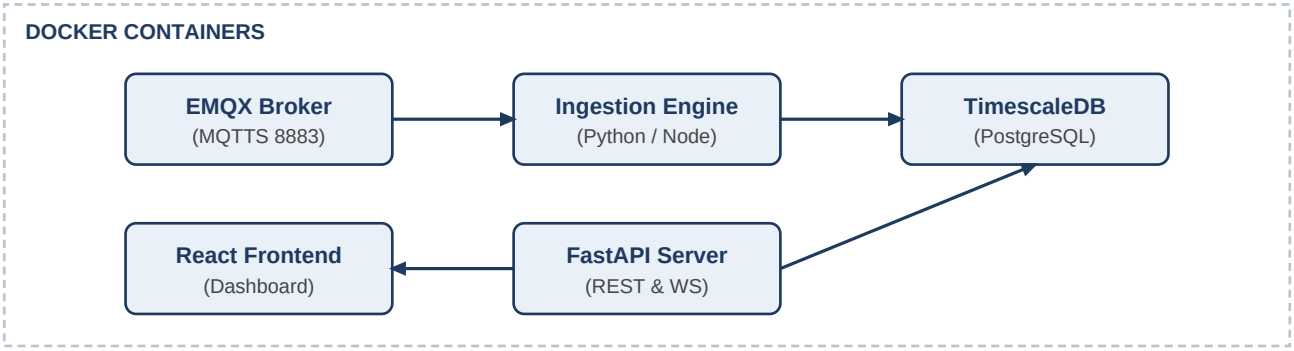
4. Power Management Subsystem (PMIC & Battery Backup)

Subsystem Element	Component Choice	Specifications & Role
Main Input Power	5V DC via USB-C / Terminal Block	Regulated 5V external input supply.
Battery Charger IC	**TP4056** or **BQ24075**	Linear Li-Ion/LiPo charger with power path management.
Backup Battery	**3.7V 18650 Li-ion Cell** (2600 mAh)	Maintains node operation during mains supply loss (at least 8 hours).
3.3V System Buck/LDO	**AMS1117-3.3 / AP2112K-3.3**	Low-noise LDO providing stable voltage to ESP32 and HX711 analog rails.
Power Loss Detect	Resistor Divider to ESP32 GPIO Interrupt	Instantly triggers power outage MQTT event prior to backup switchover.

Technical Justification

- **Seamless Power Path Switching:** The BQ24075 dynamic power-path management IC powers the system directly from the external 5V input when available while simultaneously charging the battery. When mains power fails, it seamlessly switches to battery power without resetting the ESP32 microcontroller.
- **Clean Analog Power Rail:** An ultra-low noise LDO (AP2112K) decouples micro-volt sensor bridge outputs from ESP32 digital switching noise, preventing weight calibration drift.

5. Cloud Platform & Software Stack



Stack Layer	Chosen Tool / Technology	Technical Justification
MQTT Broker	**EMQX / Eclipse Mosquitto**	Lightweight, scalable, native TLS 1.2/1.3 encryption support, dynamic user authentication, and high throughput.
Time-Series DB	**TimescaleDB** (PostgreSQL Extension)	Combines raw time-series telemetry storage (weight, temperature, sensor metrics) with full relational database features (user accounts, bin IDs, item catalogs).
Backend Framework	**FastAPI** (Python)	High-performance asynchronous API, auto-generated OpenAPI documentation, and seamless integration with Python data science/analytics libraries.
Web Dashboard	**React.js + Tailwind CSS**	Dynamic component-based rendering, WebSocket support for real-time telemetry updates, and full support for English, Sinhala, and Tamil localization.
Analytics & AI	**Python** (Pandas, Scikit-learn, Statsmodels)	Executes Holt-Winters exponential smoothing for consumption forecasting, run-out date estimation, and automated reorder alerts.
Deployment	**Docker Containers on VPS**	Standardized container deployment ensuring reproducibility across production cloud environments at low cost.